

Fractal Intelligence and the Pooling Effect: Decentering AI Design Toward Solarpunk Futures

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ABSTRACT

This position paper responds to the workshop’s central question—what it would mean to design AI in genuine harmony with nature—by proposing a structural inversion: rather than placing AI design at the center of the human–nature–technology relationship, I decenter it, treating AI as one factor among several orbiting the human. From that inverted vantage point, I develop two original concepts. First, *fractal intelligence*: not intelligence merely *inspired* by nature, but intelligence whose architecture reproduces recursive organizational principles observable across multiple biological scales—a reading that dissolves the adversarial technology-versus-nature framing the workshop identifies as a representational bias. Second, the *pooling effect*: an account of how simultaneous concentrations of attention and capital render a technology culturally load-bearing, misusable, and capable of overriding commons-based rights such as *Jokaisenoikeudet*. I situate both concepts within my practice as a worldbuilder, designer, and writer, and outline what I would contribute to the workshop’s gamified co-design activities.

CCS CONCEPTS

• **Human-centered computing** → **Participatory design; Scenario-based design**; • **Computing methodologies** → *Artificial intelligence*.

KEYWORDS

Fractal Intelligence, Pooling Effect, Biomimicry, Solarpunk, Worldbuilding, Speculative Design, Data Governance, Jokaisenoikeudet

1 BACKGROUND AND MOTIVATION

I am an alumna of The GAME School at Arizona State University, where I completed my degree in Media Arts and Sciences (Design) in December 2025. My practice sits at the junction of worldbuilding, systems design, and writing. Recent projects include *Rattle*, a speculative worldbuilding project imagining the entanglement of robotics and cattle ecologies; *Full Plates*, a brand, web, and policy project on food systems; and an ongoing essay series exploring biomimicry in social justice, emergent strategy [3], and what I call *fractal policy*—governance structures that repeat their integrity at every scale, from constitutional principle to individual case. I mentor at ASU hackathons, where I have worked to embed Principled Innovation checkpoints (Who benefits? Who may be harmed?) into rapid prototyping cycles, and I run a small optimization venture. Underneath all of this sits a design philosophy I am developing called *(un)structure*, built on three working laws: every structure holds hidden unstructure; the deeper you dissect a structure, the closer you get to essence; and creation happens where structure meets unstructure.

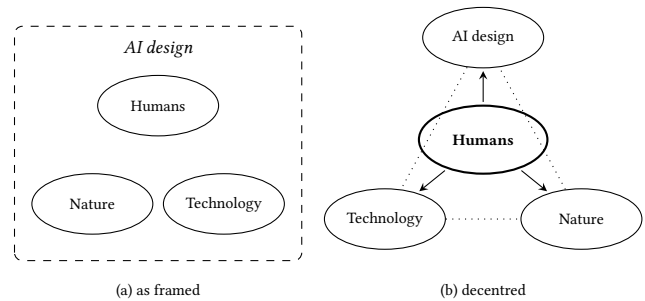


Figure 1: The inversion this paper proposes. In (a), AI design is the *context* within which humans, nature, and technology are arranged. In (b), humans are centred and AI design, technology, and nature become three relational factors—which makes the dotted relations, especially the assumed separation of AI design from technology, available for questioning.

I read the workshop’s call [1] the way I read everything—by dissecting it—and found that my annotations kept spiraling inward toward the same few questions at smaller and smaller scales. The method, I realized, was itself fractal. That recognition is the seed of this paper: the ideas below are not only *about* fractals, they were *produced* fractally, and I take that convergence of method and thesis as a sign that the workshop’s themes and my practice are genuinely aligned.

One small first reaction is worth confessing. *Jokaisenoikeudet* translates as “every person’s right,” and my margin note asked whether, in the age of AI, we should also hear it as every person’s *being right*—the right to remain an epistemic agent, to think collectively without being thought *for*. The pun is light, but the stakes underneath it are not, and they motivate everything that follows.

2 DECENTERING AI DESIGN

The workshop asks how to reimagine “the relationships between humans, nature, and technology in the context of AI design” [1]. I want to press on the grammar of that sentence. As written, AI design is the *context*—the container inside which humans, nature, and technology are arranged. My position is that we should invert the diagram: place humans at the center, and let AI, technology, and nature stand as three relational factors around them (Figure 1). This is not a cosmetic rearrangement. Once AI is a factor rather than a frame, questions become visible that the centered version quietly forecloses. Is AI independent of technology now? When did that separation happen in our collective imagination, and why?

There is precedent for this kind of move. Design theory has spent the last decade decentering the human in favor of more-than-human assemblages [6]; I am proposing the complementary operation—decentering the *artifact* that currently absorbs all our attention. And the deeper warrant is older still: Heidegger’s insistence that “the essence of technology is by no means anything technological” [7]. If the essence of AI is not in the AI, then centering AI in our design diagrams is a category error—we are staring at the surface of the structure instead of dissecting toward its essence. Heidegger’s worry was that modern technology *enframes* nature, revealing it only as standing-reserve, a stockpile of resources awaiting extraction. The workshop’s diagnosis of a “representational bias”—technology and nature rendered as adversarial, extractive, irreconcilable [1]—is, I would argue, the cultural sediment of exactly that enframing. Which raises the speculative question this paper exists to ask: what if the opposition is not merely harmful but *false*?

3 AI AS FRACTAL INTELLIGENCE

Here is my thesis, offered in the speculative register it deserves: **AI is fractal intelligence**—a recursive, self-similar replication of patterns the natural world evolved first—and it is therefore most coherently understood not as nature’s adversary but through the lens of biomimicry [2]. By fractal intelligence I do not simply mean intelligence *inspired* by nature: I mean intelligence whose architecture reproduces recursive organizational principles observable across multiple biological scales, from dendrite to network to ecosystem. Biomimicry names the design practice; fractal intelligence names the structural property that makes the practice possible.

The historical anchor for this claim is not speculative at all. The founding formal document of the field, McCulloch and Pitts’ 1943 logical calculus, is explicitly a mathematical imitation of a natural structure: the neuron [10]. Artificial intelligence did not begin as an alien intrusion into the living world; it began as an act of biomimicry, decades before Benyus gave that practice its name. Mandelbrot later showed that self-similar recursion is not an exotic curiosity but the structural grammar of nature itself—branching trees, coastlines, river networks, the dendritic architecture of the very neurons McCulloch and Pitts abstracted [9]. Neural networks are layered self-similar structures trained by recursive refinement; the lineage is not metaphorical, it is architectural. The distinction is this: a metaphor borrows an image, while an architecture inherits a mechanism. McCulloch and Pitts did not say the machine was *like* a neuron—they formalized the neuron’s threshold logic and built with it, and every layered network since carries that inheritance in its structure.

From this footing, the speculative question I would bring to the workshop’s scenario-building activities is: *what snippet of the natural world(s) is AI a replication of?* I do not think the question has one answer, and I do not want it to. Asked with a prompt-card deck in hand, it generates worlds: an AI that replicates mycorrhizal exchange behaves differently—and demands different governance—than one that replicates a murmuration, or a watershed, or a neuron. Each candidate “snippet” encodes different assumptions about ownership, access, and reciprocity. If the EdgeAI device at the center of this workshop’s ongoing research is to be experienced “in synergy

with nature rather than as an intrusion upon it,”¹ then deciding *which* natural pattern it replicates—and letting affected communities make that decision—seems to me a concrete, gameable design question rather than an abstract philosophical one. It is also where my fractal policy writing re-enters. brown’s formulation of the fractal in emergent strategy—“how we are at the small scale is how we are at the large scale”—is a governance principle before it is a metaphor [3]. Applied here, it yields a concrete rule: whatever consent standard binds the individual sensor reading must bind the aggregated dataset and the institution holding it, with no scale exempt. A device governed that way would be a small working model of *Jokaisenoikeudet* applied to computation.

4 THE POOLING EFFECT AND THE PENDULUM

Why is any of this urgent now? My answer is an original concept I call the **pooling effect**. When an economy directs enormous simultaneous concentrations of attention and capital toward a single object, that object becomes culturally load-bearing—and it scarcely matters what the object is. Load-bearing things are misusable things. Misusable things, at sufficient scale, can override commons-based rights of exactly the kind *Jokaisenoikeudet* protects. The bridge between pooled attention and eroded rights is institutional: when institutions reorganize themselves around the pooled object, previously optional participation becomes infrastructural, and meaningful consent contracts accordingly. Political ideologies pool. Entertainment franchises pool. Public figures pool. AI is pooling now, at a scale with few precedents.

The concept has a lineage I want to acknowledge rather than obscure. Simon identified the mechanism’s engine a half-century ago: in an information-rich world, attention is the scarce resource that information consumes [11]. The innovation-hype literature has mapped how expectation cycles inflate and collapse around emerging technologies [4]. What the pooling effect adds is the *rights consequence*: pooled attention and capital do not merely distort markets or expectations—they generate the political and material force to redraw what was previously held in common. The material half of this claim is already measurable. The energy and resource costs of large-scale model training are substantial and unevenly distributed [12]. And the Nordic history of *allmansrätt* shows that commons rights have always had to be actively defended against exactly such concentrations of economic interest—the pressure of commercial berry-picking on everyman’s rights is a documented case [8, 14].

The pooling effect explains why opting out of AI is no longer on the menu: the pool exists whether or not we swim. The live question is whether we can pool *deliberately*. A workshop like this one is, in effect, a counter-pool—a small concentration of attention aimed at bringing AI into equilibrium with human society and ecological limits, after the market has already made it relevant. I visualize this as a pendulum between two poles. On one side sits structured logic: the present, the economy, the referenced and reference-able. On the other sits unstructured imagination: speculative fiction, world-building, the futures we have not yet earned citations for. Neither pole alone suffices; a paper that never leaves logic changes nothing,

¹Workshop Call for Participation: <https://nordichi-solarpunk.com>

and one that never returns to it persuades no one. Solarpunk, in this scheme, is what positive worldbuilding looks like when it is anchored at *Jokaisenoikeudet*. Flynn’s manifesto names it exactly: “infrastructure as a form of resistance” [5]. The Solarpunk community has since built a shared vocabulary for futures that are ingenious rather than merely hopeful [13]. The pendulum, I note, is also the workshop’s own architecture—*Jokaisenoikeudet* as its structured legal pole, Solarpunk as its imaginative one [1]. This paper simply swings on the same hinge.

5 WHAT I WOULD BRING TO THE WORKSHOP

Four things, concretely. First, facilitation experience with gamified reflection: at ASU hackathons I introduced checkpoint prompts structurally similar to the workshop’s card typologies. Second, a worldbuilder’s fluency in moving from value statements to storyboards, exercised in *Rattle* and in my food-systems fiction. Third, a policy-writing practice ready to contribute to the research roadmap, particularly on fractal, scale-consistent governance of the EdgeAI device’s datasets. Fourth, the two concepts above, offered as prompt material: *what snippet of nature is this AI replicating?* works as a scenario card, and *what is pooling here, and what right might it override?* works as a roadmap card. I would leave the workshop with sharpened versions of both, and I would consider that a fair trade.

REFERENCES

- [1] Alan Aronica, Jamie Harper, Ed Finn, Olli Honkkila, Becky Ball, Anu Ruohomäki, Elna Ollila, and Pavan Turaga. 2026. Jokaisenoikeudet in the Age of Artificial Intelligence: Participatory Co-Design and Solarpunk Narratives in HCI. In *Adjunct Proceedings of the 14th Nordic Conference on Human-Computer Interaction (NordiCHI '26 Adjunct)*. ACM, New York, NY, USA. <https://doi.org/10.1145/3821402.3830205>
- [2] Janine M. Benyus. 1997. *Biomimicry: Innovation Inspired by Nature*. William Morrow, New York, NY.
- [3] adrienne maree brown. 2017. *Emergent Strategy: Shaping Change, Changing Worlds*. AK Press, Chico, CA.
- [4] Ozgur Dedehayir and Martin Steinert. 2016. The hype cycle model: A review and future directions. *Technological Forecasting and Social Change* 108 (2016), 28–41. <https://doi.org/10.1016/j.techfore.2016.04.005>
- [5] Adam Flynn. 2014. Solarpunk: Notes Toward a Manifesto. Hieroglyph Project, Arizona State University. Retrieved July 29, 2026 from <https://hieroglyph.asu.edu/2014/09/solarpunk-notes-toward-a-manifesto/>.
- [6] Laura Forlano. 2017. Posthumanism and Design. *She Ji: The Journal of Design, Economics, and Innovation* 3, 1 (2017), 16–29. <https://doi.org/10.1016/j.sheji.2017.08.001>
- [7] Martin Heidegger. 1977. *The Question Concerning Technology and Other Essays*. Garland Publishing, New York, NY. Translated by William Lovitt.
- [8] Matti La Mela. 2014. Property rights in conflict: wild berry-picking and the Nordic tradition of allemansrätt. *Scandinavian Economic History Review* 62, 3 (2014), 266–289. <https://doi.org/10.1080/03585522.2013.876928>
- [9] Benoit B. Mandelbrot. 1982. *The Fractal Geometry of Nature*. W. H. Freeman, San Francisco, CA.
- [10] Warren S. McCulloch and Walter Pitts. 1943. A logical calculus of the ideas immanent in nervous activity. *The Bulletin of Mathematical Biophysics* 5 (1943), 115–133. <https://doi.org/10.1007/BF02478259>
- [11] Herbert A. Simon. 1971. Designing Organizations for an Information-Rich World. In *Computers, Communications, and the Public Interest*, Martin Greenberger (Ed.). The Johns Hopkins Press, Baltimore, MD, 37–72.
- [12] Emma Strubell, Ananya Ganesh, and Andrew McCallum. 2019. Energy and Policy Considerations for Deep Learning in NLP. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*. Association for Computational Linguistics, Florence, Italy, 3645–3650. <https://doi.org/10.18653/v1/P19-1355>
- [13] The Solarpunk Community. 2018. A Solarpunk Manifesto. ReDes – Regenerative Design. Retrieved July 29, 2026 from <https://re-des.org/a-solarpunk-manifesto/>.
- [14] Pekka Tuunanen, Markus Tarasti, and Anne Rautiainen (Eds.). 2015. *Everyman’s rights and the code of conduct on private land: Existing legislation and suggestions for best practices*. Ympäristöministeriö. <https://urn.fi/URN:ISBN:978-952-11-4456-1>.